PIMS Territory Reconfiguration Portal

Google Cloud Platform (GCP) Compute Engine • Pure Bare-Metal Deployment Guide

Target Repository: https://github.com/LetscodeJosh/PIMS-Territory-Reconfig-Portal.git | **Target Branch:** main | **Environment:** GCP Compute Engine VM via PuTTY
Architecture Standard: Pure native Linux installation. Strictly zero Docker, zero containers, and zero proprietary extensions. Managed 24/7 by systemd with Nginx reverse proxy on Port 80.

1. Hardware & System Requirements

Component	Specification	Purpose & Description
Machine Type	e2-micro (Min) / e2-small (Rec)	2 vCPU, 1 GB to 2 GB RAM. Optimal for concurrent SFE planning.
Operating System	Ubuntu 22.04 LTS (or Debian 12)	Standard 64-bit Linux LTS image with long-term kernel support.
Boot Disk	10 GB to 20 GB Balanced SSD	Provides ample storage for logs, dependencies, and OS updates.
Firewall Rules	Allow HTTP (80) & HTTPS (443)	Must be enabled in GCP VPC Firewall for public browser access.
External IP	Static External IPv4	Prevents web address change when the VM stops or restarts.
Linux Packages	Python 3.10+, venv, Git, Nginx	Pure standard APT packages. Zero Docker or container dependencies.

2. Phase 1 — GCP Console Firewall Setup (One-Time)

• **Step 1.1:** Go to Google Cloud Console → *Compute Engine* → *VM instances*.
• **Step 1.2:** Click on your VM name → Click **Edit** in the top toolbar.
• **Step 1.3:** In the **Firewall** section, check both: **[x] Allow HTTP traffic** and **[x] Allow HTTPS traffic**.
• **Step 1.4:** Click **Save** at the bottom of the page.
• **Step 1.5 (Recommended):** Under *VPC network* → *IP addresses*, promote your VM IP to **Static**.

3. Phase 2 — Turnkey 1-Command Deployment via PuTTY (Fastest)

Connect to your VM via PuTTY, paste the following single command line, and press **Enter**:

```
git clone https://github.com/LetscodeJosh/PIMS-Territory-Reconfig-Portal.git ~/pims-portal && cd ~/pims-portal && bash deploy/gcp/install.sh
```

The automated script executes the following operations:
• Installs Python 3, venv, pip, git, and Nginx via standard APT.
• Creates the virtual environment (`~/pims-portal/venv`) and installs dependencies.
• Creates and activates the `pims-portal.service` systemd unit with auto-restart on boot.
• Configures Nginx reverse proxy on Port 80 with WebSocket forwarding (`_stcore/stream`).
• Tests health status and outputs your public access URL: **http://<VM_EXTERNAL_IP>**

4. Alternative: Manual Step-by-Step Commands via PuTTY

If you prefer running commands manually line by line for audit control:

Step 4.1: Install System Packages

```
sudo apt-get update -y && sudo apt-get install -y python3 python3-venv python3-pip git nginx curl
```

Step 4.2: Clone Repository

```
git clone https://github.com/LetscodeJosh/PIMS-Territory-Reconfig-Portal.git ~/pims-portal
cd ~/pims-portal
```

Step 4.3: Create Virtual Environment

```
python3 -m venv venv && source venv/bin/activate
pip install --upgrade pip && pip install -r requirements.txt
```

Step 4.4: Setup systemd Service

```
sudo cp ~/pims-portal/deploy/gcp/pims-portal.service /etc/systemd/system/pims-portal.service
sudo sed -i "s|{{USER}}|$USER|g; s|{{APP_DIR}}|$HOME/pims-portal|g;
s|{{VENV_DIR}}|$HOME/pims-portal/venv|g" /etc/systemd/system/pims-portal.service
sudo systemctl daemon-reload && sudo systemctl enable --now pims-portal.service
```

Step 4.5: Configure Nginx Reverse Proxy

```
sudo cp ~/pims-portal/deploy/gcp/nginx-pims.conf /etc/nginx/sites-available/pims-portal
sudo ln -sf /etc/nginx/sites-available/pims-portal /etc/nginx/sites-enabled/pims-portal
sudo rm -f /etc/nginx/sites-enabled/default && sudo nginx -t && sudo systemctl reload nginx
```

5. Day-to-Day Maintenance & Management Commands

Action	PuTTY Command	Description
Check Service Status	`sudo systemctl status pims-portal`	Verifies if the background service is active and running.
View Live Logs	`sudo journalctl -u pims-portal -f`	Streams real-time diagnostic output from the portal.
Restart Portal	`sudo systemctl restart pims-portal`	Safely restarts the application service.
Update to Latest Code	`cd ~/pims-portal && bash deploy/gcp/update.sh`	1-command update: pulls git commits & reloads.
Free SSL/HTTPS Setup	`bash deploy/gcp/setup_ssl.sh your-domain.com`	Installs Let's Encrypt SSL and enables HTTPS automatically.

6. Production Security & Architecture Invariants

- **Port Isolation:** Streamlit binds strictly to `127.0.0.1:8501` (loopback). Public access routes solely through Nginx on Port 80/443.
- **WebSocket Continuity:** Nginx is pre-configured with `proxy_read_timeout 86400s;` to prevent disconnects during long SFE cycle planning.
- **Zero Credential Leaks:** All system secrets are read from environment variables; no credentials exist in source files.
- **Fast Page Delivery:** Nginx Gzip compression reduces portal transfer weight by over 80%, loading in milliseconds.